Maths for EE

CBSE Class 10 2015

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

License: https://creativecommons.org/licenses/by-sa/3.0/
and
https://www.gnu.org/licenses/fdl-1.3.en.html

First manual appeared in January 2018
First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1 CBSE Class 10 Mathematics 2015 4
1.1 Formulae 4
1.2 Problems 7

1. CBSE CLASS 10 MATHEMATICS 2015

A. Formulae

Quadratic equations: For $ax^2+bx+c=0$, where $a\neq 0$, the discriminant and roots are

1.1.1

$$\Delta = b^2-4ac, \qquad x=\frac{-b\pm\sqrt{\Delta}}{2a}. \tag{1.1.1.1}$$

Equal real roots occur when $\Delta=0$. Always check that the leading coefficient is nonzero before treating an expression as a quadratic equation.

Right-triangle trigonometry and motion: For an angle θ in a right triangle,

1.1.2

$$\tan\theta=\frac{\text{opposite}}{\text{adjacent}}, \quad \sin\theta=\frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos\theta=\frac{\text{adjacent}}{\text{hypotenuse}}. \tag{1.1.2.1}$$

Useful values are $\tan 30^\circ = 1/\sqrt{3}$, $\tan 60^\circ=\sqrt{3}$, $\cot 30^\circ=\sqrt{3}$, $\cot 60^\circ = 1/\sqrt{3}$, $\sin 30^\circ = 1/2$, and $\cos 60^\circ = 1/2$. For a right triangle,

$$c^2=a^2+b^2, \qquad \|(x,y)^T\|=\sqrt{x^2+y^2}.$$

Probability: For equally likely outcomes,

1.1.3

$$P(E)=\frac{n(E)}{n(S)}. \tag{1.1.3.1}$$

When two different dice are tossed, there are $6\cdot 6=36$ equally likely ordered outcomes; exchanging the dice generally gives a different outcome.

Circles and tangents: A radius is perpendicular to the tangent at the point of contact. Tangents drawn from the same external point have equal lengths. The angle between a tangent and a chord equals the angle in the alternate segment; the angle at the centre is twice the angle at the circumference standing on the same arc. The tangent–chord theorem is

1.1.4

$$\theta_{\text{tangent}-\text{chord}}=\theta_{\text{alternate segment}}. \tag{1.1.4.1}$$

For a triangle with inradius r and semiperimeter s,

1.1.6

$$[ABC]=rs, \qquad s=\frac{AB+BC+CA}{2}. \tag{1.1.6.1}$$

Tangents from the same external point have equal lengths:

$$PA=PB. \tag{1.1.6.2}$$

Arithmetic progressions: For first term a and common difference d,

1.1.8

$$t_n=a+(n-1)d, \qquad S_n=\frac{n}{2}\big(2a+(n-1)d\big)=\frac{n}{2}(a+t_n). \tag{1.1.8.1}$$

If the unknown parameters satisfy two linear equations, write them as an augmented matrix and use Gaussian elimination (elementary row operations) to solve the system.

Coordinate geometry as linear algebra: Two nonzero vectors are perpendicular exactly when their inner product is zero. For a right angle at B,

1.1.9

$$\overrightarrow{BA}\cdot\overrightarrow{BC}=0. \tag{1.1.9.1}$$

For $A(x_1,y_1)$, $B(x_2,y_2)$, and $C(x_3,y_3)$, collinearity is equivalent to

1.1.10

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 \\ x_3-x_1 & y_3-y_1 \end{vmatrix}=0. \tag{1.1.10.1}$$

Arithmetic progressions: For an arithmetic progression,

1.1.11

$$t_n=a+(n-1)d, \qquad S_n=\frac{n}{2}\left(2a+(n-1)d\right). \tag{1.1.11.1}$$

Uniform motion obeys

1.1.13

$$\text{distance}=\text{speed}\times\text{time}, \qquad \text{average speed}=\frac{\text{total distance}}{\text{total time}}. \tag{1.1.13.1}$$

Coordinate geometry as linear algebra: Represent $P(x,y)$ by $\mathbf{p}=\begin{pmatrix}x\\y\end{pmatrix}$. Then

1.1.14

$$\overrightarrow{PQ}=\mathbf{q}-\mathbf{p}, \qquad \langle\mathbf{u},\mathbf{v}\rangle=u_xv_x+u_yv_y. \tag{1.1.14.1}$$

If P divides AB internally in the ratio $AP:PB=m:n$, then

$$\mathbf{p}=\frac{n\mathbf{a}+m\mathbf{b}}{m+n}=\mathbf{a}+\frac{m}{m+n}(\mathbf{b}-\mathbf{a}). \tag{1.1.14.2}$$

Probability: If E_1,E_2,E_3 form an exhaustive set of mutually exclusive events, then

1.1.15

$$P(E_1)+P(E_2)+P(E_3)=1. \tag{1.1.15.1}$$

Areas and solids: For a sector of angle θ degrees and radius r,

1.1.16

$$A_{\text{sector}}=\frac{\theta}{360^\circ}\pi r^2, \qquad A_{\text{segment}}=A_{\text{sector}}-A_{\text{triangle}}. \tag{1.1.16.1}$$

The curved surface areas of a cylinder and a cone are

1.1.17

$$A_{\text{cyl}}=2\pi rh, \qquad A_{\text{cone}}=\pi r\ell, \qquad \ell=\sqrt{r^2+h^2}. \tag{1.1.17.1}$$

Arithmetic progressions: The sum of terms m through n is

1.1.22

$$S_{m\ldots n}=S_n-S_{m-1}. \tag{1.1.22.1}$$

Right-triangle trigonometry and motion: For two parts of a journey, total time is

1.1.23

$$t_1+t_2=\frac{d_1}{v_1}+\frac{d_2}{v_2}. \tag{1.1.23.1}$$

For an object at fixed height h observed from level ground, its horizontal range is $x = h\cot\theta$. In this situation,

1.1.27

$$x = h\cot\theta. \tag{1.1.27.1}$$

Probability: For events A and B,

1.1.28

$$P(A\cup B) = P(A) + P(B) - P(A\cap B), \qquad P(A^c) = 1 - P(A). \tag{1.1.28.1}$$

Coordinate geometry as linear algebra: The triangle area is half the absolute determinant:

1.1.29

$$[ABC] = \frac{1}{2}\left|\begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix}\right|. \tag{1.1.29.1}$$

Areas and solids: A shaded region between an outer and an inner shape has area

1.1.30

$$A_{\text{shaded}} = A_{\text{outer}} - A_{\text{inner}}. \tag{1.1.30.1}$$

The corresponding volume formula for a cylinder is

1.1.31

$$V_{\text{cyl}} = \pi r^2 h. \tag{1.1.31.1}$$

For a hemisphere and sphere,

$$V_{\text{hemisphere}} = \frac{2}{3}\pi r^3, \qquad V_{\text{sphere}} = \frac{4}{3}\pi r^3, \qquad A_{\text{sphere}} = 4\pi r^2. \tag{1.1.31.2}$$

For a cylinder with two hemispherical cavities of radius r scooped from its ends,

$$V_{\text{remaining}} = \pi R^2 h - \frac{4}{3}\pi r^3. \tag{1.1.31.3}$$

B. Problems

1.2.1 If the quadratic equation $px^2 - 2\sqrt{5}\,px + 15 = 0$ has two equal roots, find p.
Solution: Its discriminant must be zero:

$$(-2\sqrt{5}p)^2 - 4(p)(15) = 0 \qquad (1.1.1.1)$$
$$20p^2 - 60p = 0$$
$$20p(p-3) = 0.$$

The algebra gives $p = 0$ or $p = 3$. The first value removes the quadratic and linear terms and leaves $15 = 0$, which is impossible. Therefore $p = 3$.

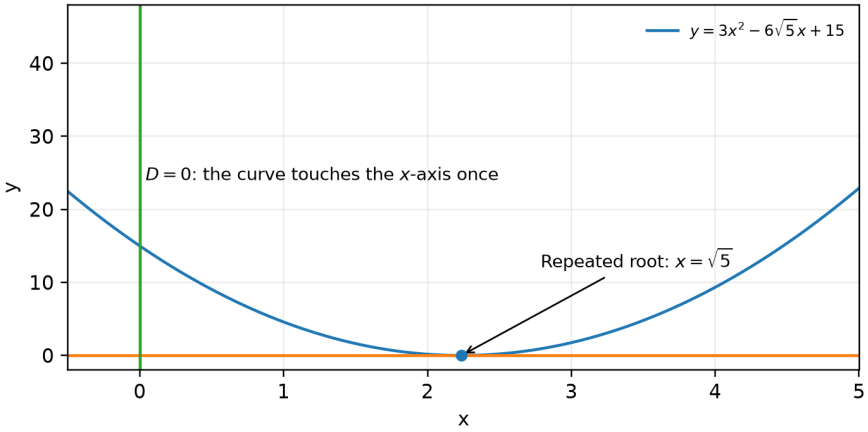


Fig. 1.2.1. For $p = 3$, the quadratic touches the x-axis at its repeated root $x = \sqrt{5}$, showing that there is only one distinct root.

The Python code uses the theoretically derived value $p = 3$ and plots its parabola:

Code reference -

codes/Q01.py

1.2.2 A tower AB is $20\,\mathrm{m}$ high and its shadow BC is $20\sqrt{3}\,\mathrm{m}$ long. Find the Sun's altitude.

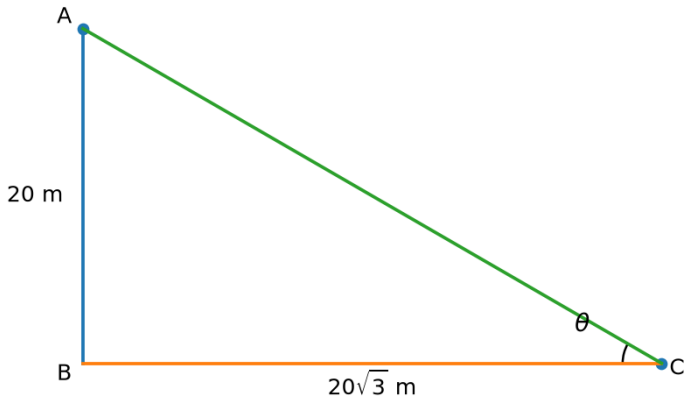


Fig. 1.2.2. Tower and its shadow.

Solution: Let the angle of elevation be θ. From the right triangle,

$$\tan\theta = \frac{AB}{BC} = \frac{20}{20\sqrt{3}} = \frac{1}{\sqrt{3}} \qquad (1.1.2.1).$$

Thus $\theta = 30°$.

The tower-and-shadow diagram is generated by this script:

Code reference -

codes/Q02.py

1.2.3 Two different dice are tossed together. Find the probability that the product of their top numbers is 6.

Solution: Each cell in the table gives the product of the ordered outcome $(\text{Die } 1, \text{Die } 2)$. There are $6 \times 6 = 36$ equally likely outcomes.

TABLE 1.2
PRODUCTS FOR ALL 36 ORDERED DICE OUTCOMES. THE FOUR PRODUCTS EQUAL TO 6 ARE HIGHLIGHTED.

Die 1 \ Die 2	1	2	3	4	5	6
1	1	2	3	4	5	**6**
2	2	4	**6**	8	10	12
3	3	**6**	9	12	15	18
4	4	8	12	16	20	24
5	5	10	15	20	25	30
6	**6**	12	18	24	30	36

The favourable outcomes are $(1,6), (2,3), (3,2), (6,1)$, so

$$P(\text{product} = 6) = \frac{4}{36} = \frac{1}{9} \qquad (1.1.3.1).$$

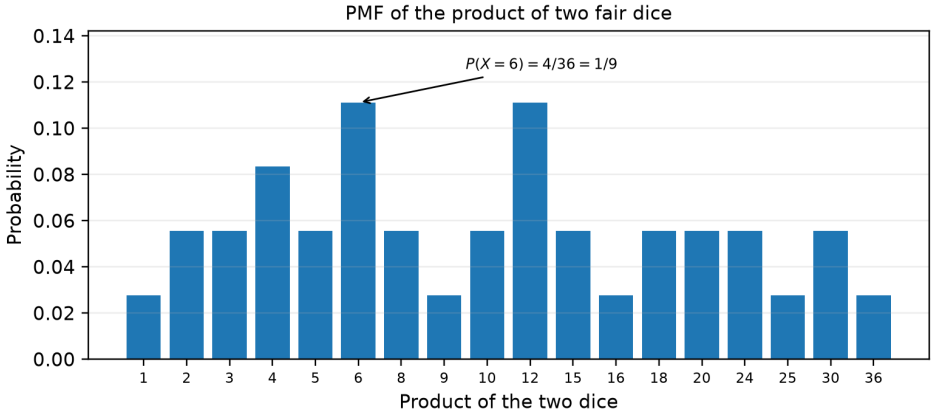


Fig. 1.2.3. Probability mass function of the product of two distinguishable fair dice. The bar at 6 represents the four favourable ordered outcomes.

The Python code uses the theoretically counted favourable outcomes for each product, divides each count by 36, and plots the probability mass function:

Code reference -

```
codes/Q03.py
```

1.2.4 PQ is a chord of a circle with centre O, and PT is a tangent. If $\angle QPT = 60°$, find $\angle PRQ$.

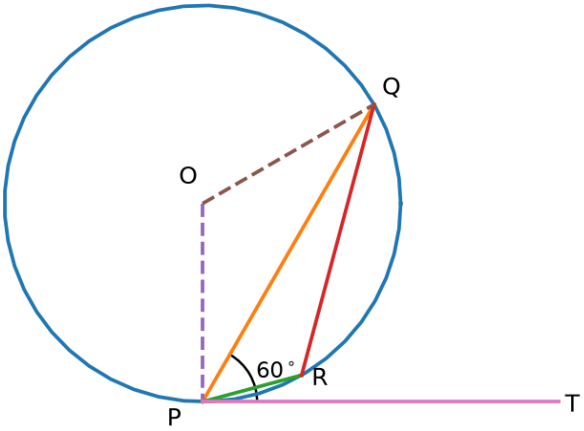


Fig. 1.2.4. Chord–tangent configuration.

Solution: Since $OP \perp PT$, we have $\angle OPQ = 30°$. As $OP = OQ$, triangle OPQ is isosceles, so $\angle OQP = 30°$ and $\angle POQ = 120°$. With R on the minor arc PQ, the intercepted major arc measures $240°$. An angle at the circumference is half its intercepted arc, hence $\angle PRQ = 120°$. The tangent–chord theorem gives the alternate-segment angle as $60°$, consistent with the supplementary angle here (1.1.4.1). The circle-and-tangent diagram is generated by this script:

Code reference -

codes/Q04.py

1.2.6 A triangle circumscribes a circle of radius $3\,\text{cm}$. The base portions are $BD = 6\,\text{cm}$ and $DC = 9\,\text{cm}$. If the triangle's area is $54\,\text{cm}^2$, find AB and AC.

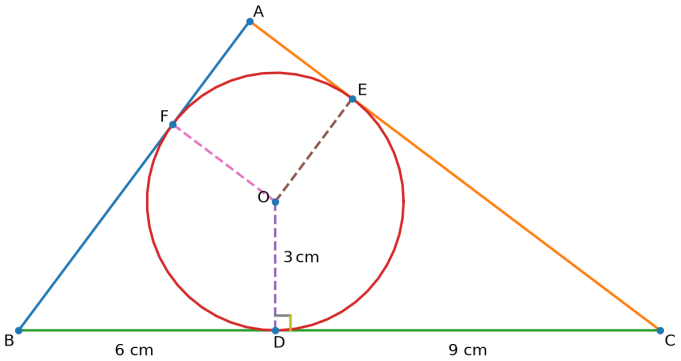


Fig. 1.2.5. Triangle with an incircle.

Solution: The semiperimeter is

$$s = \frac{[ABC]}{r} = \frac{54}{3} = 18\,\text{cm} \qquad (1.1.6.1).$$

Tangents from a common external point have equal lengths. Let the tangent length from A be x. Then $AB = x + 6$, $AC = x + 9$, and $BC = 15$ (1.1.6.2). Consequently,

$$s = \frac{(x+6) + (x+9) + 15}{2} = x + 15 = 18,$$

so $x = 3$. Therefore $AB = 9\,\text{cm}, \quad AC = 12\,\text{cm}$.
The incircle diagram is generated by this script:

Code reference -

codes/Q06.py

1.2.8 In an arithmetic progression, $S_5 + S_7 = 167$ and $S_{10} = 235$. Find the AP.
Solution: Let the first term be a and the common difference be d.

$$S_5 = 5a + 10d, \qquad S_7 = 7a + 21d, \qquad S_{10} = 10a + 45d \qquad (1.1.8.1).$$

The conditions therefore give the linear system

$$\begin{pmatrix} 12 & 31 \\ 10 & 45 \end{pmatrix} \begin{pmatrix} a \\ d \end{pmatrix} = \begin{pmatrix} 167 \\ 235 \end{pmatrix}.$$

Apply Gaussian elimination to the augmented matrix:

$$\left[\begin{array}{cc|c} 12 & 31 & 167 \\ 10 & 45 & 235 \end{array}\right] \xrightarrow{R_2 \leftarrow 6R_2 - 5R_1} \left[\begin{array}{cc|c} 12 & 31 & 167 \\ 0 & 115 & 575 \end{array}\right].$$

The second row gives $115d = 575$, so $d = 5$. Substituting this into the first row gives $12a + 31(5) = 167$, hence $a = 1$. The AP is $1, 6, 11, 16, \ldots$.
This script uses NumPy's linear solver to compare the theoretical values with the solution of the two-equation system:

Code reference -

codes/Q08.py

1.2.9 $A(4, 7)$, $B(p, 3)$ and $C(7, 3)$ form a right triangle, right-angled at B. Find p.
Solution: The two side vectors meeting at B are

$$\overrightarrow{BA} = \begin{pmatrix} 4 - p \\ 4 \end{pmatrix}, \qquad \overrightarrow{BC} = \begin{pmatrix} 7 - p \\ 0 \end{pmatrix}.$$

Since the sides are perpendicular, their inner product is zero:

$$\langle \overrightarrow{BA}, \overrightarrow{BC} \rangle = (4 - p)(7 - p) = 0 \qquad (1.1.9.1).$$

Thus $p = 4$ or $p = 7$. The second value makes $B = C$, so it does not form a triangle. Hence $p = 4$.

1.2.10 If $A(x, y)$, $B(-5, 7)$ and $C(-4, 5)$ are collinear, find the relation between x and y.
Solution: By the determinant criterion,

$$\begin{vmatrix} x + 5 & y - 7 \\ 1 & -2 \end{vmatrix} = 0 \qquad (1.1.10.1).$$

Expanding gives $-2(x + 5) - (y - 7) = 0$, and hence $2x + y + 3 = 0$.

1.2.11 The 14th term of an AP is twice its 8th term. If the 6th term is -8, find the sum of its first 20 terms.

Solution: Let a and d denote the first term and common difference. The first condition is

$$a + 13d = 2(a + 7d) \quad \Longrightarrow \quad a + d = 0 \qquad (1.1.11.1).$$

The sixth-term condition gives $a + 5d = -8$. With $a = -d$, we obtain $4d = -8$, so $d = -2$ and $a = 2$. Therefore

$$S_{20} = \frac{20}{2}\big(2(2) + 19(-2)\big) = -340 \qquad (1.1.11.1).$$

The AP terms and sum are checked by this script:

Code reference -

codes/Q11.py

1.2.13 An aeroplane is observed at an elevation of 60° from point A. After 15 seconds, the elevation becomes 30°. If its altitude remains $1500\sqrt{3}\,\text{m}$, find its speed in km/h.

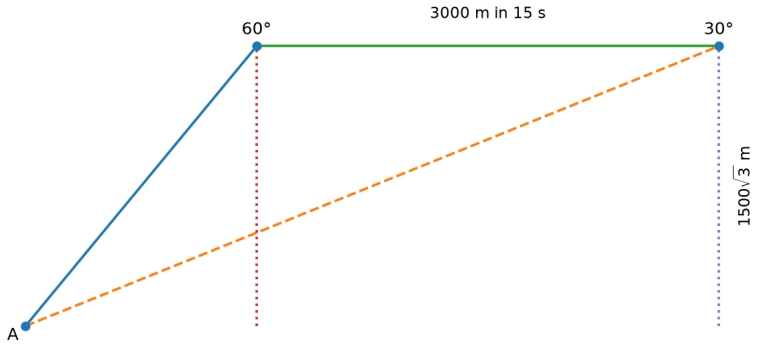


Fig. 1.2.6. The plane moves horizontally while staying at the same altitude.

Solution: Let its initial and final horizontal ranges be x_1 and x_2. Since the altitude h is fixed, $x = h\cot\theta$:

$$x_1 = 1500\sqrt{3}\cot 60^\circ = 1500\,\text{m},$$
$$x_2 = 1500\sqrt{3}\cot 30^\circ = 4500\,\text{m} \qquad (1.1.27.1).$$

The plane covers $x_2 - x_1 = 3000\,\text{m}$ in 15 s. Therefore,

$$v = \frac{3000}{15} = 200\,\text{m/s} = 200 \times \frac{18}{5} = 720\,\text{km/h} \qquad (1.1.13.1).$$

The speed calculation and accompanying figure are generated by this script:

Code reference -

codes/Q13.py

1.2.14 $A(-2,-2)$ and $B(2,-4)$. Find the point P on segment AB for which $AP = \frac{3}{7}AB$.

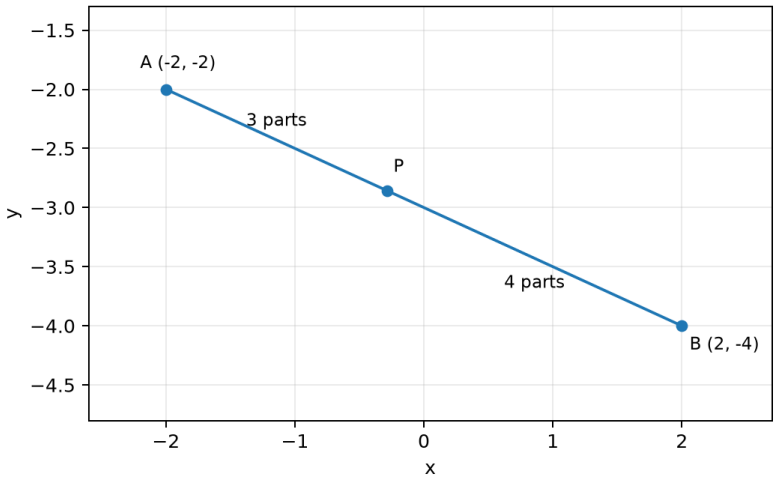


Fig. 1.2.7. The point P is three-sevenths of the way from A to B.

Solution: The section formula gives

$$\begin{aligned}\mathbf{p} &= \mathbf{a} + \frac{3}{7}(\mathbf{b}-\mathbf{a}) \qquad (1.1.14.1),\ (1.1.14.2)\\ &= \begin{pmatrix}-2\\-2\end{pmatrix} + \frac{3}{7}\begin{pmatrix}4\\-2\end{pmatrix} = \begin{pmatrix}-2/7\\-20/7\end{pmatrix}.\end{aligned}$$

Hence $P\left(-\frac{2}{7}, -\frac{20}{7}\right)$.
The section-formula diagram is generated by this script:

Code reference -

codes/Q14.py

1.2.15 A jar contains only red, blue and orange balls. The probabilities of drawing a red and a blue ball are $\frac{1}{4}$ and $\frac{1}{3}$, respectively. If there are 10 orange balls, find the total number of balls.

Solution: Let N be the total number of balls. Write each probability separately:

$$\begin{aligned}P(R) &= \frac{\text{number of red balls}}{N} = \frac{1}{4},\\ P(B) &= \frac{\text{number of blue balls}}{N} = \frac{1}{3},\\ P(O) &= \frac{\text{number of orange balls}}{N} = \frac{10}{N}.\end{aligned}$$

Since every ball is one of these three colours, $P(R)+P(B)+P(O) = 1$ (1.1.15.1).
Thus

$$\frac{1}{4} + \frac{1}{3} + \frac{10}{N} = 1 \implies \frac{10}{N} = \frac{5}{12} \implies N = 24.$$

So there are 24 balls in total: 6 red, 8 blue and 10 orange.

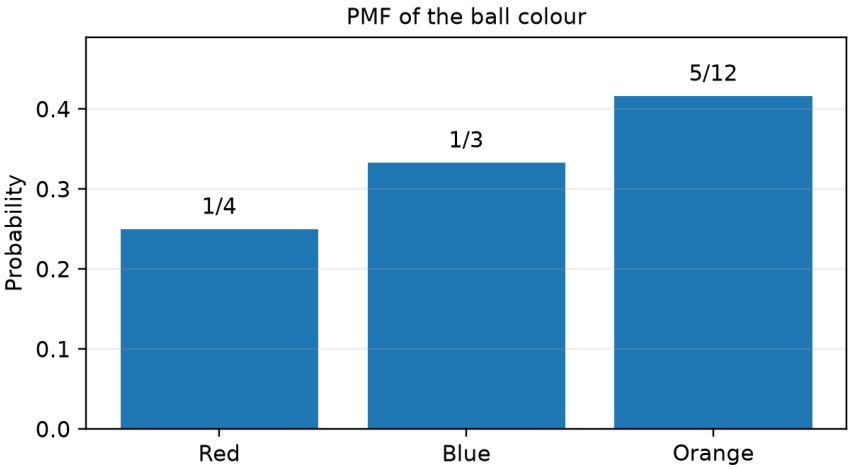


Fig. 1.2.8. Probability mass function for drawing each ball colour. The three probabilities sum to 1.

The Python code plots the theoretically derived probabilities $1/4$, $1/3$, and $5/12$ as a probability mass function:

Code reference -

codes/Q15.py

1.2.16 Find the minor and major segment areas of a circle of radius $14\,\mathrm{cm}$ cut by a chord subtending 60° at the centre. Use $\pi = \frac{22}{7}$.

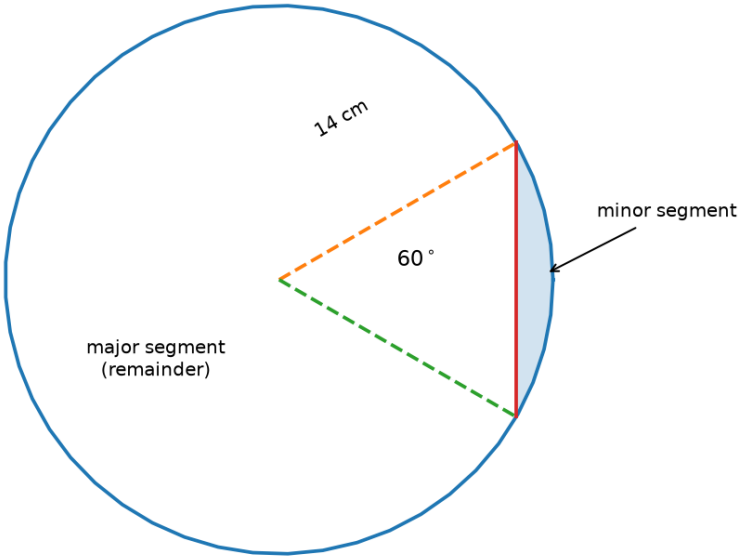


Fig. 1.2.9. The minor segment is the small cap; the major segment is the remainder of the circle.

Solution: The sector angle is 60°, so

$$A_{\text{sector}} = \frac{60}{360}\pi(14)^2 = \frac{308}{3}\,\mathrm{cm}^2 \qquad (1.1.16.1).$$

The triangle between the two radii has two sides 14 cm and included angle 60°, so it is equilateral. Its area is $\frac{\sqrt{3}}{4}(14)^2 = 49\sqrt{3}\,\mathrm{cm}^2$. Subtracting this triangle from the sector gives

$$A_{\text{minor}} = \frac{308}{3} - 49\sqrt{3}\,\mathrm{cm}^2.$$

The whole circle has area $\pi(14)^2 = 616\,\mathrm{cm}^2$. Therefore

$$A_{\text{major}} = 616 - A_{\text{minor}} = \frac{1540}{3} + 49\sqrt{3}\,\mathrm{cm}^2.$$

The segment diagram and area check are generated by this script:

Code reference -

codes/Q16.py

1.2.17 One tent has a cylindrical lower part of diameter $4.2\,\mathrm{m}$ and height $4\,\mathrm{m}$, topped by a cone with the same diameter and height $2.8\,\mathrm{m}$. Canvas costs Rs. 100 per m^2. Find the associations' 50% contribution for 100 tents, and state the values shown.

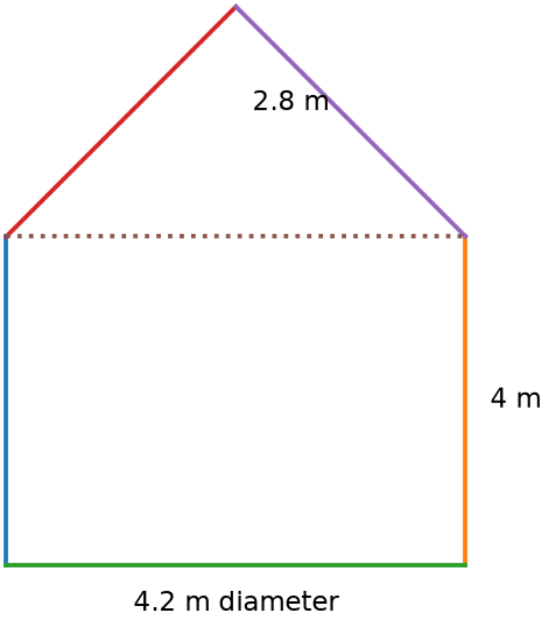


Fig. 1.2.10. The tent consists of a cylinder and a cone and has no canvas floor.

Solution: The radius is $r = 2.1\,\mathrm{m}$. The cone's slant height is

$$\ell = \sqrt{2.1^2 + 2.8^2} = 3.5\,\mathrm{m} \qquad (1.1.17.1).$$

Only the curved cylinder and cone surfaces need canvas. Their combined area is

$$\begin{aligned} A &= 2\pi rh + \pi r\ell \qquad (1.1.17.1) \\ &= 2\pi(2.1)(4) + \pi(2.1)(3.5) = 75.9\,\mathrm{m}^2. \end{aligned}$$

For 100 tents, the total cost is $100 \times 75.9 \times 100 =$ Rs. 759,000. Half is Rs. 379,500. The plan also shows cooperation, initiative and care for people affected by floods. The tent cross-section is generated by this script:

Code reference -

```
codes/Q17.py
```

1.2.22 Find the 60th term of the AP $8, 10, 12, \ldots$ and the sum of its last 10 terms.

Solution: Here $a = 8$ and $d = 2$.

$$t_{60} = 8 + 59(2) = 126 \qquad (1.1.8.1).$$

The 51st term is $t_{51} = 8 + 50(2) = 108$. The last ten terms run from 108 to 126, so

$$S_{51\ldots60} = S_{60} - S_{50} = \frac{10}{2}(108 + 126) = 1170 \qquad (1.1.22.1).$$

The 60th term and final-ten-term sum are checked by this script:

Code reference -

```
codes/Q22.py
```

1.2.23 A train travels 54 km at speed x km/h and then 63 km at $(x + 6)$ km/h. The total journey takes 3 hours. Find x.

Solution:

$$\frac{54}{x} + \frac{63}{x+6} = 3 \qquad (1.1.23.1),\ (1.1.13.1).$$

Multiplying by $x(x + 6)$ gives $54(x + 6) + 63x = 3x(x + 6)$, and hence

$$x^2 - 33x - 108 = 0 = (x - 36)(x + 3).$$

Speed must be positive, so $x = 36\,\mathrm{km/h}$.

1.2.27 Point A is 20 m above a lake. The angle of elevation of a cloud is $30°$, while the angle of depression of its reflection in the lake is $60°$. Find the distance from A to the cloud.

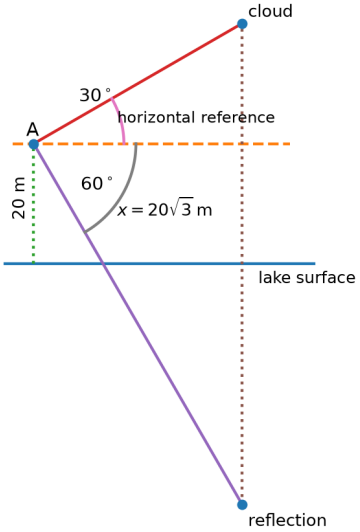


Fig. 1.2.11. The reflection is as far below the lake surface as the cloud is above it.

Solution: Let H be the cloud's height above the lake and let x be the horizontal distance from A to the vertical line through the cloud. The cloud is $H-20$ metres above A, while its reflection is $H+20$ metres below A. As shown in the figure, both the angle of elevation and the angle of depression are measured from the horizontal reference line through A. Using the two angles,

$$\frac{H-20}{x} = \tan 30^\circ = \frac{1}{\sqrt{3}},$$
$$\frac{H+20}{x} = \tan 60^\circ = \sqrt{3}. \qquad (1.1.2.1),\ (1.1.27.1)$$

Rearranging produces a linear system in the unknowns H and x:

$$\begin{pmatrix} 1 & -1/\sqrt{3} \\ 1 & -\sqrt{3} \end{pmatrix} \begin{pmatrix} H \\ x \end{pmatrix} = \begin{pmatrix} 20 \\ -20 \end{pmatrix}.$$

For clarity, solve this system by Gaussian elimination. Its augmented matrix is

$$\left[\begin{array}{cc|c} 1 & -1/\sqrt{3} & 20 \\ 1 & -\sqrt{3} & -20 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|c} 1 & -1/\sqrt{3} & 20 \\ 0 & -2/\sqrt{3} & -40 \end{array}\right].$$

The second row gives $-2x/\sqrt{3} = -40$, so $x = 20\sqrt{3}\,\mathrm{m}$. Substitution into the first row yields $H - 20 = 20$, so $H = 40\,\mathrm{m}$. Hence the vertical separation between A and the cloud is $H - 20 = 20\,\mathrm{m}$. By Pythagoras,

$$AC = \sqrt{x^2 + (H-20)^2} = \sqrt{(20\sqrt{3})^2 + 20^2} = 40\,\mathrm{m}.$$

The linear system is solved independently and the angle diagram is generated by this script:

Code reference -

`codes/Q27.py`

1.2.28 A card is drawn from a well-shuffled 52-card deck. Find the probability that it is (i) a spade or an ace, (ii) a black king, (iii) neither a jack nor a king, (iv) either a king or a queen.

Solution: Use the classical probability and union rules:

(i) The 13 spades and 4 aces overlap at the ace of spades. Thus $P = (13+4-1)/52 = 4/13$ (1.1.3.1), (1.1.28.1).

(ii) There are two black kings, so $P = 2/52 = 1/26$ (1.1.3.1).

(iii) There are four jacks and four kings. Their complement contains $52-4-4 = 44$ cards, so $P = 44/52 = 11/13$ (1.1.3.1).

(iv) Kings and queens are disjoint categories, each with four cards. Thus $P = (4+4)/52 = 2/13$ (1.1.3.1).

1.2.29 Find k if the triangle with vertices $(1,-1)$, $(-4,2k)$ and $(-k,-5)$ has area 24 square units.

Solution: Taking differences of the coordinate vectors,

$$2[ABC] = \begin{vmatrix} -5 & 2k+1 \\ -k-1 & -4 \end{vmatrix} \qquad (1.1.29.1).$$

Expanding the determinant gives

$$2[ABC] = |2k^2 + 3k + 21|.$$

Since $2k^2+3k+21$ has positive leading coefficient and discriminant $9-168 = -159 < 0$, it is positive for every real k. The specified area is 24, so the determinant's absolute value is 48 and

$$2k^2 + 3k + 21 = 48 \quad \Longrightarrow \quad 2k^2 + 3k - 27 = 0.$$

Applying the quadratic formula, the discriminant is $3^2 - 4(2)(-27) = 225$, hence

$$k = \frac{-3 \pm \sqrt{225}}{4} = \frac{-3 \pm 15}{4}.$$

Therefore both solutions are valid: $k = 3$ or $k = -\frac{9}{2}$.

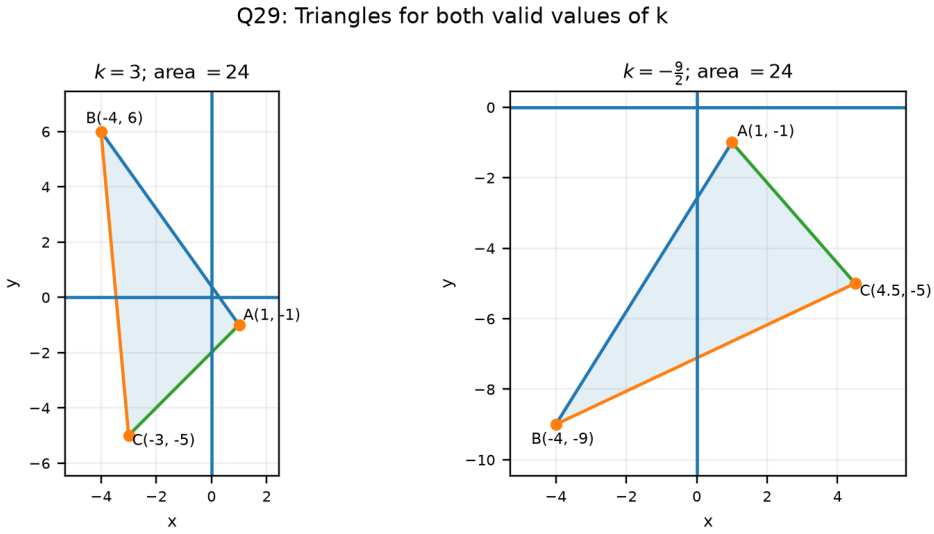


Fig. 1.2.12. The triangle formed by the given coordinates for each valid solution, $k = 3$ and $k = -\frac{9}{2}$. Both triangles have area 24 square units.

The Python code plots the triangles for the two theoretically derived values of k, using the coordinate area formula for the displayed areas:

Code reference -

codes/Q29.py

1.2.30 A square lawn has side 42 m. Two circular flower beds lie outside the vertical sides, with each circle centred at the intersection O of the square's diagonals. Find the total shaded area. Use $\pi = 22/7$.

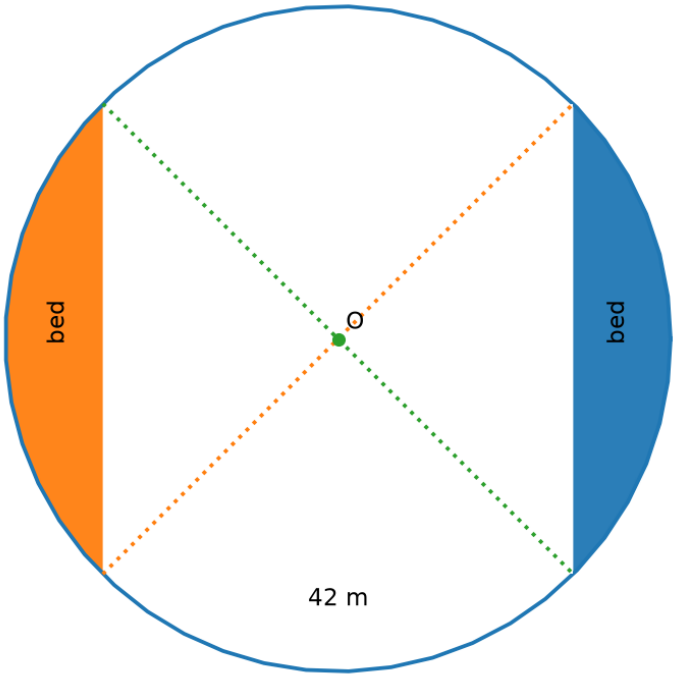


Fig. 1.2.13. The shaded parts are the two circular segments outside the square lawn.

Solution: The radius from the centre to a corner is half the diagonal. Thus

$$r^2 = \frac{42^2 + 42^2}{4} = 882\,\mathrm{m}^2.$$

Each vertical side subtends 90° at O. The area of one segment is the area of a quarter-circle sector minus the right triangle with legs 21 m and 21 m:

$$A_1 = \frac{1}{4}\pi r^2 - \frac{1}{2}(42/2)^2 = \frac{882\pi}{4} - 441.$$

The two beds are congruent, so

$$A_{\text{total}} = 2A_1 = 441\pi - 882 \qquad (1.1.30.1).$$

Using $\pi = 22/7$ gives $A_{\text{total}} = 1386 - 882 = 504\,\mathrm{m}^2$.

The flower-bed figure and shaded-area result are generated by this script:

Code reference -

codes/Q30.py

1.2.31 From both ends of a metal cylinder, hemispherical cavities of the same diameter are scooped out. The cylinder has height 10 cm and radius 4.2 cm. The remaining metal is made into a cylindrical wire of thickness 1.4 cm. Find the wire length. Use $\pi = 22/7$.

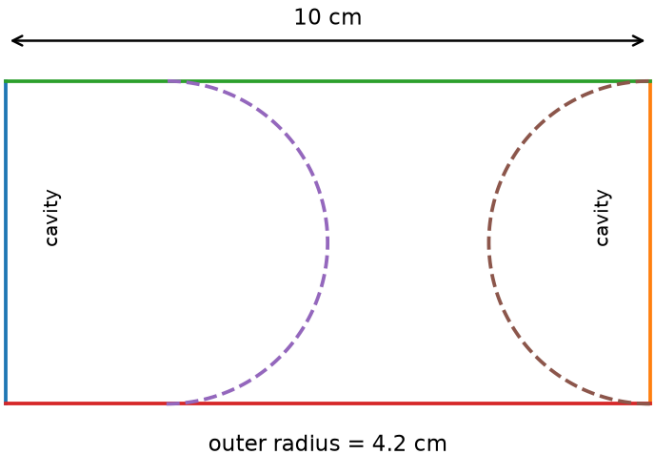


Fig. 1.2.14. Two hemispherical cavities remove the volume of one sphere.

Solution: The two hemispherical cavities together have the volume of one sphere. The remaining metal has volume

$$V = \pi(4.2)^2(10) - \frac{4}{3}\pi(4.2)^3 = 77.616\pi\,\text{cm}^3 \qquad (1.1.31.1),\ (1.1.31.2),\ (1.1.31.3).$$

The wire's thickness is its diameter, so its radius is 0.7 cm. Equating the volumes gives

$$\pi(0.7)^2 L = 77.616\pi \quad \Longrightarrow \quad L = \frac{77.616}{0.49} = 158.4\,\text{cm}.$$

The cylinder-and-cavities diagram is generated by this script:

Code reference -

codes/Q31.py